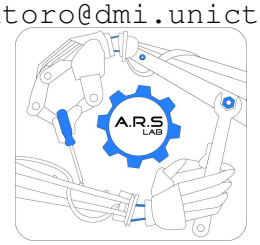
Using the 7-segments Display

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Embedded Systems

Notes

The 7-segments Display

- The Nucleo64 add-on board has a **four digit** 7-segments display
- The display can be used to show both number and letters (those that can be displayed, e.g. the "**m**" cannot be)
- Digits are numbered 0 to 3, from left to right
- The stm32_unict_lib provides functions to directly access the display
- They use TIM5, so this timer is not available when the display is enabled

Notes

Display Control Functions

- Display initialization
  `void DISPLAY_init(void)`
- Write a character “c” in digit “d”
  `void DISPLAY_putc(int d, char c)`
- Write a string “s” from digit “d”
  `void DISPLAY_puts(int digit, char * s)`
- Turn on or off the decimal point of digit “d”
  `void DISPLAY_dp(int digit, int on_off)`

Notes

Example: tenth-seconds counter

```
#include "stm32_unict_lib.h"
#include <stdio.h>

int counter = 0;

int main()
{
    // Init the display
    DISPLAY_init();

    // init the timer 2 and configure the timebase
    // tick set to 0.1 ms
    TIM_init(TIM2);
    TIM_config_timebase(TIM2, 8400, 1000);
    TIM_set(TIM2, 0);
    TIM_on(TIM2);
    TIM_enable_irq(TIM2, IRQ_UPDATE);

    for (;;) {
        char s[5];
        sprintf(s,"%4d", counter); // put counter value to string
        DISPLAY_puts(0, s); // update the display
    }
}

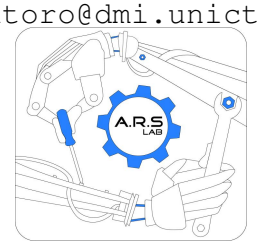
void TIM2_IRQHandler(void)
{
    ++counter;
    TIM_update_clear(TIM2);
}
```


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